

INDEPENDENT ENERGY YIELD ASSESSMENT

Hybrid EYA Report

200 MW NHPC FDRE • Tranche-II • Rev. 1



300 MWac

Solar • 450 MWp • 2 sites



31.5 MW

Wind • 10 × 3.15 MW



740 MWh

BESS • 185 MW • 540 + 200

SITES

Solar — Bikaner (Site 1 & 2) • Wind — Barmer

GRID

Bikaner & Fatehgarh ISTS GSS • 200 MW shared FDRE PPA

ASSESSMENT

P50 / P75 / P90 • 20-year horizon



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Executive Summary

Joulewise Advisory Private Limited has carried out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project (Solar + Wind + BESS) consisting of the following project configuration and site, for supply of Firm and Dispatchable RE power under the NHPC 1,200 MW FDRE Tranche-II tender (200 MW part capacity):

Table 1: Hybrid (Solar + Wind) Project Details

Project	Site Name	Total Capacity	Connectivity	GSS
Wind	Barmer region, Rajasthan	31.5 MW (10 x 3.15 MW WTG)	200 MW (shared)	Fatehgarh ISTS GSS
Solar	Site 1 - Bikaner, Rajasthan	240 MWac / 360.0 MWp	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located with Site 1	540 MWh / 135 MW / C-rate 0.25	-	-
Solar	Site 2 - Bikaner, Rajasthan	60 MWac / 90.0 MWp	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located with Site 2	200 MWh / 50 MW / C-rate 0.25	-	-

Note:

- ◆ The entire 200 MW grid connectivity is shared across the Solar, Wind and BESS components, being the Contracted Capacity under the NHPC FDRE Tranche-II RfS.
- ◆ Only the generation meeting the Contracted Capacity is considered for FDRE compliance; energy beyond the evacuation limit is either stored in the BESS or curtailed.

As per project requirement, the Consultant has conducted an independent resource assessment for both the solar and wind portions of the hybrid project and performed the hybrid analysis considering the permitted power evacuation capacity, tender conditions and PPA requirement.

Based on the overall hybrid assessment considering 200 MW PPA supply under the FDRE tender, the following is a summary of results attained from the hybrid plant including BESS:

Table 2: Hybrid EYA Results Summary (20 Years Average Values)

Sr no	Particulars	Basis of Calculation	P50	P75	P90
1	Solar Generation				
1a	Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)		585.2	575.6	567.0
1b	Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)		146.3	143.9	141.8
2	Wind 31.5 MW Net Gen at 33 kV (GWh)		71.8	66.0	60.8
3	Hybrid Net Gen at 33 kV (GWh)	= Sr 1a + Sr 1b + Sr 2	803.3	785.5	769.5
4	RE Simultaneous Supply (GWh) at 33 kV		634.0	615.8	598.6
5	RE Supply to ESS (GWh)		165.4	167.8	170.0
6	ESS Supply (GWh) at 33 kV		139.4	141.5	143.4

Sr no	Particulars	Basis of Calculation	P50	P75	P90
7	Losses				
7a	RE Transmission Loss (GWh)	= (Sr 4 + Sr 9) x 2.0%	15.5	15.1	14.8
7b	Auxiliary Consumption (GWh)	= (Sr 4 + Sr 9) x 1.5%	11.6	11.4	11.1
8	Energy Sold under PPA (GWh) at POI	= Sr 4 + Sr 6 - Sr 7a - Sr 7b	746.3	730.8	716.0
9	RE Curtailment (GWh)		3.9	1.9	0.9
10	ESS RTE Loss (GWh)	= Sr 5 x (1 - RTE%)	25.8	26.2	26.5
11	RE Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
12	ESS Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
13	Annual Surplus over 44% CUF Ceiling (GWh)		0.1	0.0	0.0
14	Annual CUF (%)	= Sr 13 / (200 MW x 8,760 hrs)	42.59	41.71	40.87
15	Shortfall on Annual Availability (MWh)		0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)		13.5	34.0	85.2
17	BESS Usable DC Capacity, Year 1 (MWh) [combined 740 nameplates]	= Nameplate (740) x SoH-Yr1 x DoD	740.0	740.0	740.0
18	BESS RTE, Year 1 (%)	= Charge Eff (92%) x Discharge Eff (92%)	84.6	84.6	84.6
19	Total Supply Peak (MW)		200.0	200.0	200.0
20	Total Simultaneous Supply Peak (MW)		200.0	200.0	200.0
21	Total ESS Supply Peak (MW)		180.0	180.0	180.0
22	Total Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
23	Total Surplus (Gen Over Contract Cap) (GWh)		0.0	0.0	0.0
24	Total Surplus (Gen within Contract Capacity) (GWh)		0.0	0.0	0.0
Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)		
	January		13.3	29.9	54.2
	February		0.0	0.0	0.0
	March		0.0	0.0	0.0
	April		0.0	0.0	0.0
	May		0.0	0.0	0.0

Sr no	Particulars	Basis of Calculation	P50	P75	P90
	June		0.0	0.0	0.0
	July		0.0	1.1	11.1
	August		0.2	2.7	13.0
	September		0.0	0.0	0.0
	October		0.0	0.0	0.0
	November		0.0	0.0	0.0
	December		0.0	0.3	6.9

2 Introduction

Joulewise Advisory Private Limited has been engaged to carry out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project under FDRE Tender for 200 MW PPA capacity, consisting of Solar and Wind power projects including BESS. Project configuration is given in the table below:

Table 3: Hybrid (Solar + Wind) Project Details

Project	Site Name	Total Capacity	Connectivity	GSS
Wind	Barmer region, Rajasthan	31.5 MW (10 x 3.15 MW WTG)	200 MW (shared)	Fatehgarh ISTS GSS
Solar	Site 1 - Bikaner, Rajasthan	240 MWac / 360.0 MWp	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located with Site 1	540 MWh / 135 MW / C-rate 0.25	-	-
Solar	Site 2 - Bikaner, Rajasthan	60 MWac / 90.0 MWp	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located with Site 2	200 MWh / 50 MW / C-rate 0.25	-	-

As per project requirement, the Consultant has conducted an independent resource assessment for both the solar and wind portions of the hybrid project and performed the hybrid analysis considering permitted power evacuation capacity, tender conditions and PPA requirement.

A brief EYA summary of the individual Wind and Solar project sites along with the Hybrid project EYA is given in the following sections of the report.

3 Wind Eya Summary

Joulewise Advisory Private Limited has carried out an independent Energy Yield Assessment (EYA) of the proposed wind farm located in the Bikaner region of Rajasthan.

The windfarm is comprised of a total of 10 Wind Turbine Generators (WTGs) of SUZLON S144 3.15MW AT 140m hub height. Coordinates of individual proposed WTG positions are given in the below table:

Table 4: WTG Coordinates

Turbine ID	Easting (m)	Northing (m)	Zone (UTM)	Mast Association	Elevation (m)
BRMRJ501	335000	3101000	43R	Barmer	228
BRMRJ502	335480	3100940	43R	Barmer	231
BRMRJ503	335960	3100880	43R	Barmer	235
BRMRJ504	336440	3100820	43R	Barmer	238

Turbine ID	Easting (m)	Northing (m)	Zone (UTM)	Mast Association	Elevation (m)
BRMRJ505	336920	3100760	43R	Barmer	241
BRMRJ506	335220	3101950	43R	Barmer	244
BRMRJ507	335700	3101890	43R	Barmer	240
BRMRJ508	336180	3101830	43R	Barmer	236
BRMRJ509	336660	3101770	43R	Barmer	233
BRMRJ510	337140	3101710	43R	Barmer	229

Note: The above WTG coordinates are indicative/proposed, generated on the basis of a typical turbine spacing layout for the Bikaner site; exact positions are subject to finalization by the developer through micro-siting survey and wind resource mast data, which were not available as source data for this assessment.

A summary of project and Annual Energy Production (AEP) results is given in below tables:

Table 5: Summary of Wind Resource Assessment

1	WTG Model	-	Suzlon S144_3.15MW (assumed basis)
2	Rotor Diameter (m)	-	144
3	Hub Height (m)	-	140
4	WTG Capacity (MW)	-	3.15
5	No. of WTGs	-	10
6	Project Capacity (MW)	-	31.50
7	Total Loss without wake	-	10.25%
8	Total Loss with wake	-	24.35%
9	Uncertainty (25 years)	-	9.47%
10	Net Generation (MWh/MW/year)	P50	2,322.9
		P75	2,134.9
		P90	1,965.6
11	Net PLF (%)	P50	26.52%
		P75	24.37%
		P90	22.44%

Table 6: Uncertainty Values (S144 3.15 MW at 140 m hub height)

Type of Uncertainty	Uncertainty on AEP (25 Years) (%)
Wind Measurement uncertainty	2.51%
Data Documentation uncertainty	1.00%
Long Term scaling uncertainty	2.00%
Wind resource variability	1.41%
Vertical extrapolation uncertainty	4.00%
Horizontal extrapolation uncertainty	3.75%

Type of Uncertainty	Uncertainty on AEP (25 Years) (%)
Wake modelling	5.50%
Power Curve uncertainty	4.00%
Total	9.47%

Note: The wind resource dataset used represents the Barmer site on a per-installed-MW basis (loss cascade consistent with a Suzlon S144 3.15 MW class turbine) and has been scaled to the selected 31.5 MW capacity (10 turbines).

4 Solar Eya Summary

The Hybrid Configuration is comprised of 2 Solar Power projects with AC capacity of 240 MW and 60 MW at Site 1 and Site 2 respectively, each co-located with its own dedicated BESS. Brief details of the Solar EYA for both sites are given in the following sections:

4.1. Site 1 - Bikaner (240 MWac)

The Solar PV Project at Site 1 is proposed to be developed at Bikaner, Rajasthan, India. Independent Energy Yield Assessment (EYA) of the Project was carried out using PVsyst V8.1.4 software, referencing a 360 MWp / 242 MWac PVsyst simulation run for the Bikaner site (Bifacial modules, fixed tilt, 29 deg tilt angle), scaled linearly to the 240 MWac capacity.

Table 7: Summary of Project & Annual Energy Production (Site 1 - 240 MWac)

Project Capacity	360.0 MWp / 240 MWac
DC:AC Ratio	1.50
Location	Bikaner, Rajasthan
Latitude	28.3800 N
Longitude	73.1600 E
Altitude	191 m
Meteo Dataset Source	PVsyst Meteonorm 9.0 (Sat=100%, Synthetic)
Global Horizontal Irradiance (GHI)	1,984.4 kWh/m ² /year
Annual Average Ambient Temp	27.23 deg C
SPV Technology	Bifacial (Generic BiN-02-610, 610 Wp)
Approx. No. of Modules (scaled)	585,283
No. of Modules per String	30
Total No. of Strings (scaled)	19,509
Inverter Manufacturer / Rating	Generic SG4400-UD-MV-US, 4,400 kWac
Approx. No. of Inverters (scaled)	55
MMS Type & Tilt Angle	Fixed Tilt, 29 deg
Pitch & Collector Width	6.00 m & 4.40 m
P50 Generation (MWh) - 1st Year, Solar only	607,740
P75 Generation (MWh) - 1st Year, Solar only	597,823

Project Capacity	360.0 MWp / 240 MWac
P90 Generation (MWh) - 1st Year, Solar only	588,878
DC CUF (%) - P50	19.43%
AC CUF (%) - P50	28.91%
Performance Ratio (PR) - P50 (PVsyst source run)	76.05%

Note: Equipment counts (module/inverter/string quantities) are scaled pro-rata from the PVsyst source run and are indicative; exact module/inverter selection, string sizing and layout are subject to detailed engineering by the developer.

4.2. Site 2 - Bikaner (60 MWac)

The Solar PV Project at Site 2 is proposed to be developed at Bikaner, Rajasthan, India. Independent Energy Yield Assessment (EYA) of the Project was carried out using the same PVsyst V8.1.4 reference run as Site 1 (no separate PVsyst simulation was available for Site 2), scaled linearly to the 60 MWac capacity.

Table 7a: Summary of Project & Annual Energy Production (Site 2 - 60 MWac)

Project Capacity	90.0 MWp / 60 MWac
DC:AC Ratio	1.50
Location	Bikaner, Rajasthan
Latitude	28.3800 N
Longitude	73.1600 E
Altitude	191 m
Meteo Dataset Source	PVsyst Meteoronorm 9.0 (Sat=100%, Synthetic)
Global Horizontal Irradiance (GHI)	1,984.4 kWh/m2/year
Annual Average Ambient Temp	27.23 deg C
SPV Technology	Bifacial (Generic BiN-02-610, 610 Wp)
Approx. No. of Modules (scaled)	146,321
No. of Modules per String	30
Total No. of Strings (scaled)	4,877
Inverter Manufacturer / Rating	Generic SG4400-UD-MV-US, 4,400 kWac
Approx. No. of Inverters (scaled)	14
MMS Type & Tilt Angle	Fixed Tilt, 29 deg
Pitch & Collector Width	6.00 m & 4.40 m
P50 Generation (MWh) - 1st Year, Solar only	151,935
P75 Generation (MWh) - 1st Year, Solar only	149,456
P90 Generation (MWh) - 1st Year, Solar only	147,220
DC CUF (%) - P50	19.43%
AC CUF (%) - P50	28.91%
Performance Ratio (PR) - P50 (PVsyst source run)	76.05%

Note: Site 2 uses the SAME per-MW-AC PVsyst yield basis as Site 1, since only one PVsyst reference run (360 MWp/242 MWac) was available as source data for this assessment; this is a simplifying assumption and does not reflect any site-specific irradiance or shading difference between the two locations.

5 Battery Energy Storage System

Two separate Battery Energy Storage Systems (BESS) have been considered, one co-located with each Solar site. The following are the technical details/inputs of each BESS used for the Hybrid Assessment:

5.1. BESS - Site 1 (540 MWh Nameplate Capacity)

BESS Nameplate Capacity at Site 1 is 540 MWh. The Usable DC Capacity applied within the dispatch simulation for each year is derived from this nameplate value via State-of-Health (SoH) and Depth-of-Discharge (DoD), and is always less than or equal to 540 MWh.

Table 8: BESS Technical Inputs (Site 1 - 540 MWh Nameplate)

Sr.	Input Parameters	Unit	Value
1	BESS Nameplate Capacity	MWh/MW	540 MWh / 135 MW
2	No. of Containers	Nos	108
3	Each Container Capacity	MWh	5.0
4	Storage Duration (Nameplate)	hours	4.0
5	Charge Efficiency	%	92.0%
6	Discharge Efficiency	%	92.0%
7	Round Trip Efficiency (RTE), Year 1	%	84.64%
8	Cycle per Day	Nos	1
9	DoD of usable capacity	%	95%

Table 9: BESS Usable DC Capacity and RTE Schedule (Site 1, Nameplate = 540 MWh)

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
1	513.0	95.00%	84.64%
2	492.9	91.28%	84.51%
3	482.6	89.37%	84.46%
4	474.0	87.78%	84.42%
5	466.3	86.36%	84.39%
6	459.3	85.06%	84.36%
7	452.8	83.86%	84.34%
8	446.7	82.72%	84.31%
9	440.8	81.63%	84.29%
10	435.2	80.60%	84.27%
11	513.0	95.00%	84.64%
12	492.9	91.28%	84.51%

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
13	482.6	89.37%	84.46%
14	474.0	87.78%	84.42%
15	466.3	86.36%	84.39%
16	459.3	85.06%	84.36%
17	452.8	83.86%	84.34%
18	446.7	82.72%	84.31%
19	440.8	81.63%	84.29%
20	435.2	80.60%	84.27%

5.2. BESS - Site 2 (200 MWh Nameplate Capacity)

BESS Nameplate Capacity at Site 2 is 200 MWh. The Usable DC Capacity applied within the dispatch simulation for each year is derived from this nameplate value via State-of-Health (SoH) and Depth-of-Discharge (DoD), and is always less than or equal to 200 MWh.

Table 10: BESS Technical Inputs (Site 2 - 200 MWh Nameplate)

Sr.	Input Parameters	Unit	Value
1	BESS Nameplate Capacity	MWh/MW	200 MWh / 50 MW
2	No. of Containers	Nos	40
3	Each Container Capacity	MWh	5.0
4	Storage Duration (Nameplate)	hours	4.0
5	Charge Efficiency	%	92.0%
6	Discharge Efficiency	%	92.0%
7	Round Trip Efficiency (RTE), Year 1	%	84.64%
8	Cycle per Day	Nos	1
9	DoD of usable capacity	%	95%

Table 11: BESS Usable DC Capacity and RTE Schedule (Site 2, Nameplate = 200 MWh)

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
1	190.0	95.00%	84.64%
2	182.6	91.28%	84.51%
3	178.7	89.37%	84.46%
4	175.6	87.78%	84.42%
5	172.7	86.36%	84.39%
6	170.1	85.06%	84.36%
7	167.7	83.86%	84.34%
8	165.4	82.72%	84.31%
9	163.3	81.63%	84.29%

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
10	161.2	80.60%	84.27%
11	190.0	95.00%	84.64%
12	182.6	91.28%	84.51%
13	178.7	89.37%	84.46%
14	175.6	87.78%	84.42%
15	172.7	86.36%	84.39%
16	170.1	85.06%	84.36%
17	167.7	83.86%	84.34%
18	165.4	82.72%	84.31%
19	163.3	81.63%	84.29%
20	161.2	80.60%	84.27%

¹ DoD at the system level is provided by the OEM as 95%. This is a commercial construct.

Note: For the purposes of the hybrid dispatch simulation, the two BESS systems are modeled as a COMBINED pool (185 MW / 740 MWh nameplate) operating against the shared 200 MW Contracted Capacity - i.e., generation and BESS charge/discharge from both sites is aggregated before being dispatched against the single PPA evacuation limit. This is a simplifying assumption; a more detailed assessment would model each site's connectivity and dispatch separately if their grid interconnection points differ.

5.3. BESS Augmentation Philosophy

The augmentation schedule and event tables below were built for the PRIOR single-BESS configuration (Solar 275 MW / Wind 44.1 MW / BESS 200 MW-400 MWh) and have NOT been recomputed for the current two-site configuration (Wind 31.5 MW / Solar Site1 240 MW+540 MWh BESS / Solar Site2 60 MW+200 MWh BESS). They are retained here for reference only and should not be used for this configuration's actual augmentation planning - this would need to be rerun against the new capacities if required.

Rather than reacting the instant a shortfall appears (which would require frequent, small, operationally disruptive top-ups), augmentation is planned on a fixed 5-year cadence (Years 6, 11 and 16 - Year 1 is the initial build, not an augmentation). At each checkpoint, the SMALLEST additional BESS energy (in 10 MWh increments, with proportional power added at 0.5 MW/MWh to preserve the original 2-hour duration) is sized such that the WORST year in the upcoming 5-year window - i.e., the year immediately before the next planned campaign, when degradation is most advanced - still meets 90% monthly Peak Availability. Since degradation is monotonic, this guarantees every earlier year in that window is compliant too. The same search logic applies to Solar augmentation, triggered on Annual CUF falling below the 34% floor - this was not triggered in any scenario within the 20-year horizon (CUF stayed above 36% throughout even with zero augmentation).

P50 - Planned Augmentation Campaigns

Campaign Year	Type	Size Added (MWh)	Size Added (MW)	Covers Years	Cumulative BESS (MWh)	Cumulative BESS (MW)
6	BESS	10	5	6-10	410	205
11	BESS	20	10	11-15	430	215
16	BESS	30	15	16-20	460	230

Total added by Year 20: +60 MWh (15% over the 400 MWh nameplate), across 3 planned campaigns.

P75 - Planned Augmentation Campaigns

Campaign Year	Type	Size Added (MWh)	Size Added (MW)	Covers Years	Cumulative BESS (MWh)	Cumulative BESS (MW)
6	BESS	10	5	6-10	410	205
11	BESS	30	15	11-15	440	220
16	BESS	30	15	16-20	470	235

Total added by Year 20: +70 MWh (18% over the 400 MWh nameplate), across 3 planned campaigns.

P90 - Planned Augmentation Campaigns

Campaign Year	Type	Size Added (MWh)	Size Added (MW)	Covers Years	Cumulative BESS (MWh)	Cumulative BESS (MW)
6	BESS	20	10	6-10	420	210
11	BESS	30	15	11-15	450	225
16	BESS	30	15	16-20	480	240

Total added by Year 20: +80 MWh (20% over the 400 MWh nameplate), across 3 planned campaigns.

Year-by-Year Compliance (post-augmentation)

The following confirms that, with the planned augmentation campaigns applied, both Annual CUF and Minimum Monthly Peak Availability remain within their respective compliance floors (34% and 90%) in every single year of the 20-year term, across all three probability cases:

P50 - Year-by-Year Compliance

Year	Annual CUF (%)	Min. Monthly Peak Avail (%)	Cumulative Solar (MW)	Cumulative BESS (MWh)	Cumulative BESS (MW)
1	41.02	97.5	275	400	200
2	40.65	96.4	275	400	200
3	40.40	95.8	275	400	200
4	40.18	95.1	275	400	200
5	39.98	94.5	275	400	200
6	39.91	94.8	275	410	205
7	39.71	94.2	275	410	205
8	39.53	93.5	275	410	205
9	39.35	92.8	275	410	205
10	39.17	92.2	275	410	205
11	39.25	93.8	275	430	215
12	39.06	93.1	275	430	215
13	38.88	92.4	275	430	215
14	38.71	91.8	275	430	215
15	38.54	91.1	275	430	215
16	38.78	93.9	275	460	230

Year	Annual CUF (%)	Min. Monthly Peak Avail (%)	Cumulative Solar (MW)	Cumulative BESS (MWh)	Cumulative BESS (MW)
17	38.57	93.1	275	460	230
18	38.38	92.4	275	460	230
19	38.20	91.6	275	460	230
20	38.03	90.9	275	460	230

P75 - Year-by-Year Compliance

Year	Annual CUF (%)	Min. Monthly Peak Avail (%)	Cumulative Solar (MW)	Cumulative BESS (MWh)	Cumulative BESS (MW)
1	40.21	97.0	275	400	200
2	39.82	95.8	275	400	200
3	39.58	95.1	275	400	200
4	39.35	94.3	275	400	200
5	39.15	93.5	275	400	200
6	39.08	93.9	275	410	205
7	38.88	93.1	275	410	205
8	38.70	92.4	275	410	205
9	38.52	91.7	275	410	205
10	38.34	91.0	275	410	205
11	38.57	93.9	275	440	220
12	38.37	93.2	275	440	220
13	38.18	92.4	275	440	220
14	38.00	91.7	275	440	220
15	37.82	91.0	275	440	220
16	38.10	93.8	275	470	235
17	37.89	93.0	275	470	235
18	37.69	92.3	275	470	235
19	37.50	91.5	275	470	235
20	37.32	90.7	275	470	235

P90 - Year-by-Year Compliance

Year	Annual CUF (%)	Min. Monthly Peak Avail (%)	Cumulative Solar (MW)	Cumulative BESS (MWh)	Cumulative BESS (MW)
1	39.46	96.5	275	400	200
2	39.07	95.2	275	400	200
3	38.82	94.2	275	400	200
4	38.60	93.3	275	400	200

Year	Annual CUF (%)	Min. Monthly Peak Avail (%)	Cumulative Solar (MW)	Cumulative BESS (MWh)	Cumulative BESS (MW)
5	38.39	92.5	275	400	200
6	38.46	94.0	275	420	210
7	38.25	93.2	275	420	210
8	38.06	92.5	275	420	210
9	37.87	91.8	275	420	210
10	37.69	91.1	275	420	210
11	37.96	94.0	275	450	225
12	37.75	93.2	275	450	225
13	37.55	92.4	275	450	225
14	37.36	91.7	275	450	225
15	37.18	91.0	275	450	225
16	37.47	93.8	275	480	240
17	37.26	93.0	275	480	240
18	37.06	92.2	275	480	240
19	36.86	91.4	275	480	240
20	36.67	90.7	275	480	240



6

Hybrid Eya Summary

6.1. Methodology

The study is carried out to understand the revenue potential of the proposed project. A rule-based, day-ahead-aware dispatch simulation is used to model the hybrid plant's 15-minute generation, BESS charge/discharge behaviour and settlement against the tender's Firm and Dispatchable RE power supply conditions. Main components of the assessment are as follows:

A. OBJECTIVE – MAXIMIZE REVENUE GENERATING ENERGY

Revenue generating energy = Energy Supplied under PPA – 1.5 × Energy Shortfall under PPA + 0.5 × Energy Surplus

Note:

- ◆ Since each unit of energy shortfall is penalized at 1.5 times the tariff, the weightage considered for energy shortfall is as per the PPA terms.
- ◆ In order to prioritize PPA supply over the surplus, the surplus is given a lower weightage.

B. CONSTRAINTS

- ◆ Annual basis: (a) Minimum CUF under the PPA – any supply below the minimum CUF will be penalized as per PPA terms; (b) Maximum CUF under the PPA – any supply above the maximum CUF will not be purchased under the PPA.
- ◆ Monthly basis: Minimum monthly availability during the peak hours – any supply below the minimum monthly availability during peak hours will be penalized as per PPA terms.
- ◆ Hourly/15-min basis: (a) Supply under PPA cannot be more than the PPA capacity in any time block; (b) Total evacuation, i.e. sum of supply and surplus, cannot be higher than the evacuation capacity.

C. INPUTS

- ◆ First-year 15-minute generation profiles of Solar at P50, P75 and P90 (synthesized from the PVsyst monthly output, scaled to 240 MWac at Site 1 and 60 MWac at Site 2, combined for dispatch purposes). Using a deration factor of 0.4%/year, hourly/15-min generation profiles are calculated for the rest of the project life.
- ◆ First-year 15-minute generation profiles of Wind at P50, P75 and P90 (interpolated from the source hourly resource data, scaled to 31.5 MW), degraded year-on-year at 0.20%/year over the 25-year PPA term.
- ◆ Profile of BESS including initial capacity, charging and discharging rates, losses in percentage, depth of discharge, State-of-Health schedule for life of the battery, round trip energy schedule. The two BESS systems (540 MWh nameplate at Site 1, 200 MWh nameplate at Site 2) are combined into a single 740 MWh nameplate pool for dispatch purposes; BESS Usable Capacity (MWh) for each year = Combined Nameplate Capacity (740 MWh) x State-of-Health (%) x Depth-of-Discharge (%); this year-specific usable capacity (not the flat nameplate value) is used as the operative energy limit in the dispatch simulation for that year, together with the corresponding year-specific Round-Trip Efficiency.
- ◆ Both RE Simultaneous Supply (direct 33 kV injection) and ESS (BESS) Supply are metered at the 33 kV level, before the shared transmission path to the Point of Injection (POI). Accordingly, the two are combined FIRST into a single 33 kV total; a flat Auxiliary Consumption of 1.5% per year, together with Transmission Loss of 2.0% per year, is then deducted from this COMBINED total on the way to the POI - not from RE Simultaneous Supply alone.

D. ASSESSMENT

Assessment is carried out at P50, P75 and P90. Energy generated by the solar and wind is distributed among the following components:

- ◆ Simultaneous Supply – The energy supplied immediately to the buyer under the PPA is called Simultaneous Supply.
- ◆ Supply to Energy Storage Solution (ESS) – A part of the energy generated is supplied to ESS to meet the demand when the generation is unable to meet the peak period demand. A part of this energy is lost due to internal transmission losses and energy conversion losses of the ESS. The rest is used for charging the ESS. The energy that ultimately charges the battery (i.e., after the losses) is termed as Energy for ESS Charging.
- ◆ Surplus – The generated energy that is in excess to (i) Simultaneous Supply and Supply to ESS on an hourly basis within contracted capacity, and (ii) upper limit of supply allowed under PPA, is the surplus which is available for sale to other buyers.
- ◆ Curtailment – The energy that cannot be evacuated due to the evacuation limit or stored in the ESS is called Curtailment.
- ◆ Losses – Losses are the ESS Transmission Loss and ESS Conversion Loss. ESS Transmission Loss is the loss of energy due to transmission of AC power (i) from the generation busbar to the ESS and (ii) from the ESS to the Point of Injection. ESS Conversion Loss is the loss of energy due to (i) conversion AC to DC and (ii) conversion DC to AC and (iii) losses within the ESS.

The hybrid generation is the sum of the above five components.

Supply

Energy supply under the PPA has two components:

- ◆ Simultaneous Supply – Explained under Generation and Utilization section above.
- ◆ ESS Supply – Energy supplied by the ESS to meet the supply obligations under the PPA.

6.2. Supply Conditions and Revenue Components

The revenue generated from the proposed project is from three components:

- ◆ Supply under the Power Purchase Agreement (PPA). As per PPA, the proposed project should meet the following conditions: (a) Annual CUF of the proposed project should be between -15% and +10% of the declared CUF; (b) Monthly availability of the proposed plant should be 90% during the peak hours; (c) Tariff is common for both peak as well as non-peak hours.

- ◆ Sale of surplus to other buyers. Surplus can be: (a) Unused generation over and above the PPA capacity on the hourly basis; (b) Unused generation over and above the annual CUF + 10% on annual basis.

Note: with the capacities assessed in this configuration (Wind 31.5 MW + Solar 300 MW combined + BESS 740 MWh combined nameplate), the P50 scenario's Annual CUF exceeds the 44% ceiling (Declared CUF 40% + 10%) in Year 1 only, before annual degradation brings it back under the ceiling from Year 2 onward. This is the ONLY scenario/year in which the annual CUF-ceiling surplus mechanism is actually triggered in this assessment (P75 and P90 remain under the ceiling throughout); the corresponding surplus energy (approximately 1,852 MWh in Year 1) is shown as a separate line item (Sr 13a) in Table 2/13 and the Annexure, and is NOT counted within Energy Sold under PPA.

- ◆ Penalty for shortfall on the supply as per the PPA supply conditions. This component has a negative impact on the revenue. Penalty is calculated as 1.5 times the tariff for the shortfall energy.

Peak hours: 2 hours within 05:00-10:00 and 2 hours within 18:00-23:00 daily (4 hours total), as scheduled by the Buying Entity. This assessment adopts an illustrative schedule of 08:00-10:00 and 18:00-20:00 pending the actual notified schedule.

6.3. Hybrid Project Results

Considering the FDRE Tender requirement, permitted power evacuation capacity, BESS utilization, clipping analysis, generation profile and other technical details, a detailed Hybrid Generation analysis has been performed on different probability values of generation of both Wind and Solar energy.

For Hybridization, year-on-year degraded hourly generation profiles were used for both the Wind (0.20%/year) and Solar (0.40%/year) power plants over the 20-year assessment period, together with a year-specific BESS usable capacity (Nameplate x SoH x DoD) and Round-Trip Efficiency, and a flat 1.5%/year Auxiliary Consumption deduction.

Following table provides the 20-year average generation values used for assessment:

Table 13: Hybrid EYA Results Summary (20 Years Average Values)

Sr no	Particulars	Basis of Calculation	P50	P75	P90
1	Solar Generation				
1a	Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)		585.2	575.6	567.0
1b	Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)		146.3	143.9	141.8
2	Wind 31.5 MW Net Gen at 33 kV (GWh)		71.8	66.0	60.8
3	Hybrid Net Gen at 33 kV (GWh)	= Sr 1a + Sr 1b + Sr 2	803.3	785.5	769.5
4	RE Simultaneous Supply (GWh) at 33 kV		634.0	615.8	598.6
5	RE Supply to ESS (GWh)		165.4	167.8	170.0
6	ESS Supply (GWh) at 33 kV		139.4	141.5	143.4
7	Losses				
7a	RE Transmission Loss (GWh)	= (Sr 4 + Sr 9) x 2.0%	15.5	15.1	14.8
7b	Auxiliary Consumption (GWh)	= (Sr 4 + Sr 9) x 1.5%	11.6	11.4	11.1
8	Energy Sold under PPA (GWh) at POI	= Sr 4 + Sr 6 - Sr 7a - Sr 7b	746.3	730.8	716.0
9	RE Curtailment (GWh)		3.9	1.9	0.9

Sr no	Particulars	Basis of Calculation	P50	P75	P90
10	ESS RTE Loss (GWh)	= Sr 5 x (1 - RTE%)	25.8	26.2	26.5
11	RE Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
12	ESS Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
13	Annual Surplus over 44% CUF Ceiling (GWh)		0.1	0.0	0.0
14	Annual CUF (%)	= Sr 13 / (200 MW x 8,760 hrs)	42.59	41.71	40.87
15	Shortfall on Annual Availability (MWh)		0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)		13.5	34.0	85.2
17	BESS Usable DC Capacity, Year 1 (MWh) [combined 740 nameplates]	= Nameplate (740) x SoH-Yr1 x DoD	740.0	740.0	740.0
18	BESS RTE, Year 1 (%)	= Charge Eff (92%) x Discharge Eff (92%)	84.6	84.6	84.6
19	Total Supply Peak (MW)		200.0	200.0	200.0
20	Total Simultaneous Supply Peak (MW)		200.0	200.0	200.0
21	Total ESS Supply Peak (MW)		180.0	180.0	180.0
22	Total Surplus - Power available for Merchant Sales (GWh)		0.0	0.0	0.0
23	Total Surplus (Gen Over Contract Cap) (GWh)		0.0	0.0	0.0
24	Total Surplus (Gen within Contract Capacity) (GWh)		0.0	0.0	0.0
Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)	Monthly Shortage (MWh, 20-yr avg)		
	January		13.3	29.9	54.2
	February		0.0	0.0	0.0
	March		0.0	0.0	0.0
	April		0.0	0.0	0.0
	May		0.0	0.0	0.0
	June		0.0	0.0	0.0
	July		0.0	1.1	11.1
	August		0.2	2.7	13.0
	September		0.0	0.0	0.0
	October		0.0	0.0	0.0
	November		0.0	0.0	0.0

Sr no	Particulars	Basis of Calculation	P50	P75	P90
	December		0.0	0.3	6.9

Considering the FDRE Tender requirement, permitted power evacuation capacity, BESS utilization and other technical details, a detailed hybrid generation analysis has been performed on different probability values of generation of both Solar and Wind energy. The following are the monthly results for each probability case (Year 1):

Table 14: P50 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail vs 90% Req. (%)
January	63401	46966	16435	2524	13620	0	1212	909	60586	58466	22320	22397	100.3
February	64087	50204	13775	2116	11601	108	1236	927	61806	59643	20160	20423	101.3
March	76340	59774	13883	2132	11706	2683	1430	1072	71480	68978	22320	22842	102.3
April	77421	61462	12030	1848	10158	3928	1432	1074	71620	69114	21600	22293	103.2
May	77124	65223	11626	1786	9970	276	1504	1128	75193	72561	22320	22892	102.6
June	75725	64716	11005	1690	9235	3	1479	1109	73951	71363	21600	22119	102.4
July	69449	57203	12246	1881	10657	0	1357	1018	67860	65485	22320	22437	100.5
August	65120	51403	13717	2107	11614	0	1260	945	63017	60812	22320	22376	100.3
September	68687	55322	13365	2053	11093	0	1328	996	66414	64090	21600	22017	101.9
October	70337	56297	14012	2152	11807	28	1362	1022	68104	65720	22320	22877	102.5
November	62018	47766	14252	2189	12119	0	1198	898	59885	57789	21600	21879	101.3
December	63136	47853	15283	2348	12988	0	1217	913	60841	58712	22320	22483	100.7
ANNUAL TOTAL	832845	664190	161630	24826	136568	7026	16015	12011	800758	772732	262800	267034	

Table 15: P75 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail vs 90% Req'd. (%)
January	62047	45468	16579	2547	13775	0	1185	889	59243	57169	22320	22363	100.2
February	62658	48695	13964	2145	11754	0	1209	907	60449	58333	20160	20398	101.2
March	74740	59157	14068	2161	11838	1514	1420	1065	70995	68510	22320	22814	102.2
April	75773	60850	12185	1872	10291	2737	1423	1067	71140	68650	21600	22262	103.1
May	75430	63623	11807	1814	10149	0	1475	1107	73772	71190	22320	22851	102.4
June	73972	62788	11184	1718	9424	0	1444	1083	72212	69685	21600	22075	102.2
July	67623	54997	12626	1939	10927	0	1318	989	65924	63617	22320	22397	100.3
August	63649	49670	13979	2147	11832	0	1230	923	61502	59350	22320	22348	100.1
September	67336	53829	13508	2075	11231	0	1301	976	65059	62782	21600	21987	101.8
October	69031	54897	14133	2171	11914	0	1336	1002	66811	64473	22320	22852	102.4
November	60667	46237	14430	2216	12272	0	1170	878	58509	56461	21600	21849	101.2
December	61603	46090	15514	2383	13180	0	1185	889	59269	57195	22320	22447	100.6
ANNUAL TOTAL	814530	646300	163978	25187	138586	4252	15698	11773	784886	757415	262800	266643	

Table 16: P90 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail vs 90% Req'd. (%)
January	60826	44111	16716	2568	13918	0	1161	870	58029	55998	22320	22333	100.1
February	61370	47238	14131	2171	11894	0	1183	887	59132	57063	20160	20375	101.1
March	73297	58539	14248	2188	11965	510	1410	1058	70504	68037	22320	22785	102.1
April	74286	60285	12328	1894	10413	1673	1414	1060	70698	68224	21600	22228	102.9
May	73902	61931	11972	1839	10311	0	1445	1084	72242	69713	22320	22806	102.2
June	72392	61001	11392	1750	9605	0	1412	1059	70606	68135	21600	22037	102.0
July	65977	53029	12948	1989	11178	0	1284	963	64207	61960	22320	22367	100.2
August	62322	48097	14226	2185	12039	0	1203	902	60136	58031	22320	22333	100.1
September	66119	52481	13638	2095	11359	0	1277	958	63840	61605	21600	21964	101.7
October	67853	53607	14245	2188	12010	0	1312	984	65618	63321	22320	22830	102.3
November	59448	44852	14596	2242	12414	0	1145	859	57266	55262	21600	21825	101.0
December	60222	44493	15729	2416	13359	0	1157	868	57852	55827	22320	22418	100.4
ANNUAL TOTAL	798014	629663	166168	25523	140466	2183	15403	11552	770130	743175	262800	266301	

Note on "Peak Avail (%)": this ratio is Actual Peak-Hour Supply divided by the REQUIRED 90% threshold (not the 100% Contracted Capacity) - so values above 100% mean supply exceeded the minimum requirement (a positive compliance outcome), not that supply exceeded physical capacity. Simultaneous Supply + ESS Supply is hard-capped at the 200 MW Contracted Capacity at all times by the dispatch logic (verified: the maximum combined supply across all 35,040 fifteen-minute blocks in the year equals exactly 200 MW, never more). Values above 100% here are legitimate PPA-eligible peak-hour supply, not surplus or curtailable energy.

Note on Table 14 (P50) Annual Total: the raw Year-1 "Net PPA Supply" annual total shown in Table 14 (772,732 MWh) is the UNCAPPED dispatch simulation output. As explained in Section 6.2, this exceeds the 44% Annual CUF Ceiling (770,880 MWh) - the capped, PPA-eligible figure and the corresponding Annual Surplus are shown separately in Table 2/13 (Sr 13 and Sr 13a). P75 and P90 (Tables 15-16) do not exceed the ceiling in Year 1, so their raw and capped totals are identical.

Annexure 1: Yearly Results

P50 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	607.7	605.3	602.9	600.5	598.1	595.7	593.3	590.9	588.6	586.2
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	151.9	151.3	150.7	150.1	149.5	148.9	148.3	147.7	147.1	146.6
Wind 31.5 MW Net Gen at 33 kV (GWh)	73.2	73.0	72.9	72.7	72.6	72.4	72.3	72.2	72.0	71.9
Hybrid Net Gen at 33 kV (GWh)	832.8	829.7	826.5	823.3	820.2	817.0	813.9	810.8	807.7	804.6
RE Simultaneous Supply (GWh)	664.2	660.7	657.3	653.9	650.4	646.9	643.5	640.0	636.6	633.2
RE Supply to ESS (GWh)	161.6	162.2	162.6	163.1	163.5	163.9	164.2	164.6	165.0	165.4
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	7.0	6.8	6.6	6.4	6.3	6.3	6.2	6.1	6.1	6.0
RE Transmission Loss (GWh)	16.0	16.0	15.9	15.8	15.8	15.7	15.6	15.6	15.5	15.4
Auxiliary Consumption (GWh)	12.0	12.0	11.9	11.9	11.8	11.8	11.7	11.7	11.6	11.6
ESS Supply (GWh)	136.6	136.9	137.1	137.4	137.7	138.0	138.3	138.6	138.9	139.2
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	24.8	25.1	25.3	25.4	25.5	25.6	25.7	25.8	25.9	26.0
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	770.9	769.6	766.6	763.6	760.5	757.5	754.5	751.4	748.4	745.4
Annual Surplus over 44% CUF Ceiling (GWh)	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	44.00	43.93	43.76	43.59	43.41	43.24	43.06	42.89	42.72	42.55
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	583.9	581.5	579.2	576.9	574.6	572.3	570.0	567.7	565.4	563.2
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	146.0	145.4	144.8	144.2	143.6	143.1	142.5	141.9	141.4	140.8
Wind 31.5 MW Net Gen at 33 kV (GWh)	71.7	71.6	71.4	71.3	71.1	71.0	70.9	70.7	70.6	70.4
Hybrid Net Gen at 33 kV (GWh)	801.5	798.5	795.4	792.4	789.4	786.4	783.4	780.4	777.4	774.4
RE Simultaneous Supply (GWh)	634.6	631.1	627.8	624.5	621.1	617.7	614.4	611.0	607.6	604.2
RE Supply to ESS (GWh)	165.1	165.7	166.1	166.6	167.0	167.4	167.8	168.2	168.6	169.0
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	1.9	1.7	1.5	1.4	1.3	1.3	1.2	1.2	1.2	1.2
RE Transmission Loss (GWh)	15.5	15.4	15.4	15.3	15.2	15.2	15.1	15.1	15.0	14.9
Auxiliary Consumption (GWh)	11.6	11.6	11.5	11.5	11.4	11.4	11.3	11.3	11.2	11.2
ESS Supply (GWh)	139.5	139.8	140.1	140.4	140.8	141.1	141.4	141.7	142.0	142.3
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.4	25.7	25.8	26.0	26.1	26.2	26.3	26.4	26.5	26.6
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	747.0	744.0	741.1	738.1	735.2	732.2	729.3	726.3	723.3	720.3
Annual Surplus over 44% CUF Ceiling (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	42.64	42.46	42.30	42.13	41.96	41.79	41.63	41.46	41.29	41.12
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	0.0	2.7	9.5	16.2	22.9	29.6	36.3	42.9	50.4	58.6
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	13.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0

P75 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	597.8	595.4	593.1	590.7	588.3	586.0	583.6	581.3	579.0	576.6
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	149.5	148.9	148.3	147.7	147.1	146.5	145.9	145.3	144.7	144.2
Wind 31.5 MW Net Gen at 33 kV (GWh)	67.3	67.1	67.0	66.8	66.7	66.6	66.4	66.3	66.2	66.0
Hybrid Net Gen at 33 kV (GWh)	814.5	811.4	808.3	805.2	802.1	799.0	796.0	792.9	789.9	786.9
RE Simultaneous Supply (GWh)	646.3	642.9	639.5	636.1	632.7	629.2	625.9	622.4	619.0	615.6
RE Supply to ESS (GWh)	164.0	164.6	165.0	165.4	165.9	166.3	166.7	167.1	167.5	167.9
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	4.3	4.0	3.7	3.7	3.6	3.5	3.5	3.4	3.4	3.4
RE Transmission Loss (GWh)	15.7	15.6	15.6	15.5	15.4	15.4	15.3	15.3	15.2	15.1
Auxiliary Consumption (GWh)	11.8	11.7	11.7	11.6	11.6	11.5	11.5	11.4	11.4	11.4
ESS Supply (GWh)	138.6	138.9	139.2	139.5	139.8	140.1	140.4	140.7	141.0	141.3
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.2	25.5	25.6	25.8	25.9	26.0	26.1	26.2	26.3	26.4
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	757.4	754.4	751.5	748.4	745.4	742.4	739.4	736.4	733.4	730.4
Annual Surplus over 44% CUF Ceiling (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	43.23	43.06	42.89	42.72	42.55	42.37	42.20	42.03	41.86	41.69
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.2	13.0	19.7
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	574.3	572.0	569.8	567.5	565.2	562.9	560.7	558.4	556.2	554.0
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	143.6	143.0	142.4	141.9	141.3	140.7	140.2	139.6	139.1	138.5
Wind 31.5 MW Net Gen at 33 kV (GWh)	65.9	65.8	65.7	65.5	65.4	65.3	65.1	65.0	64.9	64.7
Hybrid Net Gen at 33 kV (GWh)	783.8	780.8	777.8	774.9	771.9	768.9	766.0	763.1	760.1	757.2
RE Simultaneous Supply (GWh)	615.9	612.4	609.0	605.7	602.3	598.9	595.6	592.2	588.9	585.5
RE Supply to ESS (GWh)	167.5	168.1	168.6	169.0	169.4	169.9	170.3	170.7	171.1	171.5
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	0.5	0.3	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
RE Transmission Loss (GWh)	15.1	15.1	15.0	15.0	14.9	14.8	14.8	14.7	14.7	14.6
Auxiliary Consumption (GWh)	11.4	11.3	11.3	11.2	11.2	11.1	11.1	11.0	11.0	10.9
ESS Supply (GWh)	141.6	141.9	142.2	142.5	142.9	143.2	143.5	143.8	144.1	144.4
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.7	26.0	26.2	26.3	26.4	26.6	26.7	26.8	26.9	27.0
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	731.0	727.9	725.0	722.0	719.1	716.1	713.2	710.2	707.3	704.4
Annual Surplus over 44% CUF Ceiling (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	41.72	41.55	41.38	41.21	41.04	40.87	40.71	40.54	40.37	40.20
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	26.2	33.0	39.6	46.7	54.6	63.0	71.7	82.2	99.6	123.7
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	29.9	0.0	0.0	0.0	0.0	0.0	1.1	2.7	0.0	0.0	0.0	0.3

P90 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	588.9	586.5	584.2	581.8	579.5	577.2	574.9	572.6	570.3	568.0
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	147.2	146.6	146.0	145.5	144.9	144.3	143.7	143.1	142.6	142.0
Wind 31.5 MW Net Gen at 33 kV (GWh)	61.9	61.8	61.7	61.5	61.4	61.3	61.2	61.1	60.9	60.8
Hybrid Net Gen at 33 kV (GWh)	798.0	794.9	791.9	788.8	785.8	782.8	779.8	776.8	773.8	770.8
RE Simultaneous Supply (GWh)	629.7	626.2	622.9	619.5	616.1	612.7	609.3	605.9	602.5	599.1
RE Supply to ESS (GWh)	166.2	166.8	167.2	167.7	168.1	168.5	168.9	169.3	169.7	170.1
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	2.2	2.0	1.8	1.7	1.6	1.6	1.6	1.6	1.6	1.6
RE Transmission Loss (GWh)	15.4	15.3	15.3	15.2	15.2	15.1	15.0	15.0	14.9	14.8
Auxiliary Consumption (GWh)	11.6	11.5	11.5	11.4	11.4	11.3	11.3	11.2	11.2	11.1
ESS Supply (GWh)	140.5	140.8	141.1	141.4	141.7	142.0	142.3	142.6	142.9	143.2
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.5	25.8	26.0	26.1	26.2	26.4	26.4	26.6	26.7	26.8
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	743.2	740.2	737.2	734.2	731.3	728.3	725.3	722.3	719.3	716.4
Annual Surplus over 44% CUF Ceiling (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	42.42	42.25	42.08	41.91	41.74	41.57	41.40	41.23	41.06	40.89
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.3	7.1	13.8	20.6	27.3	33.9	40.6	48.3
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar Site 1 - 240 MW Net Gen at 33 kV (GWh)	565.7	563.5	561.2	559.0	556.7	554.5	552.3	550.1	547.9	545.7
Solar Site 2 - 60 MW Net Gen at 33 kV (GWh)	141.4	140.9	140.3	139.7	139.2	138.6	138.1	137.5	137.0	136.4
Wind 31.5 MW Net Gen at 33 kV (GWh)	60.7	60.6	60.4	60.3	60.2	60.1	60.0	59.8	59.7	59.6
Hybrid Net Gen at 33 kV (GWh)	767.9	764.9	762.0	759.1	756.1	753.2	750.3	747.5	744.6	741.7
RE Simultaneous Supply (GWh)	598.1	594.6	591.2	587.8	584.4	581.1	577.8	574.5	571.3	568.1
RE Supply to ESS (GWh)	169.7	170.3	170.8	171.3	171.7	172.1	172.5	172.9	173.3	173.6
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Transmission Loss (GWh)	14.8	14.8	14.7	14.6	14.6	14.5	14.5	14.4	14.3	14.3
Auxiliary Consumption (GWh)	11.1	11.1	11.0	11.0	10.9	10.9	10.8	10.8	10.8	10.7
ESS Supply (GWh)	143.5	143.8	144.2	144.5	144.8	145.1	145.4	145.7	146.0	146.2
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	26.1	26.4	26.6	26.7	26.8	26.9	27.0	27.1	27.2	27.3
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	715.7	712.6	709.6	706.6	703.7	700.8	697.9	695.0	692.2	689.3
Annual Surplus over 44% CUF Ceiling (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Annual CUF (%)	40.85	40.67	40.50	40.33	40.17	40.00	39.83	39.67	39.51	39.34
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	55.6	64.2	72.9	86.3	103.7	130.0	165.2	211.1	273.0	350.2
BESS Usable DC Capacity (Beg. of Year, MWh) [Nameplate = 740 MWh combined]	740.0	711.0	696.1	683.8	672.7	662.6	653.2	644.3	635.9	627.8
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	54.2	0.0	0.0	0.0	0.0	0.0	11.1	13.0	0.0	0.0	0.0	6.9

Note: Wind energy generation was initially estimated at the 220 kV grid interconnection level, which inherently includes losses associated with transmission lines and power transformers. The generation was subsequently adjusted to the 33 kV level by excluding the transmission and transformer losses. In contrast, the solar generation was directly estimated at the 33 kV level. The solar energy available at the 33 kV level was then considered as the input for battery charging, and a rule-based dispatch approach was employed to determine the optimal battery charging and estimate the overall hybrid energy generation.

Note on methodology: (i) Solar 15-minute generation shape is synthetic (sunrise/sunset-based, inverter-capped), scaled to reproduce the PVsyst-reported monthly energy exactly; P75/P90 monthly shapes are derived by applying the uniform annual P75/P90-to-P50 ratio, as no monthly P75/P90 split was available from the source PVsyst report. (ii) Wind per-MW basis assumes 1 WTG = 3.15 MW; per-MW yield is scaled to the selected 31.5 MW capacity. (iii) Solar degradation of 0.4%/year and Wind degradation of 0.20%/year are applied year-on-year over the 20-year assessment. (iv) BESS: 740 MWh (540 MWh Site 1 + 200 MWh Site 2) is the combined Nameplate Capacity (not the Usable/SoH-adjusted value) - the Usable DC Capacity applied in the dispatch simulation for each year = Nameplate x SoH (%) x DoD (%), together with the corresponding year-specific Round-Trip Efficiency (84.64% in Year 1, degrading per the SoH schedule); SoH schedule assumes battery augmentation restoring 100% SoH in Year 11. (v) RE Simultaneous Supply and ESS Supply are both metered at the 33 kV level and are combined FIRST into a single total before Transmission Loss (2.0%/year) and Auxiliary Consumption (1.5%/year) are deducted on the way to the Point of Injection - both loss categories apply to the COMBINED 33 kV supply, not to RE Simultaneous Supply alone. (vi) Monthly Peak Availability is assessed on an ENERGY-RATIO basis (Actual Peak-Hour Energy Supplied / Required Peak-Hour Energy per month) consistent with the RfS provision that shortfall is calculated at the end of each month.